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TaiChi-AQA: A Dataset and Framework for Action Quality Assessment and Visual Analysis

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ABSTRACT

Action Quality Assessment (AQA) has become an advanced technology applied in various domains. However, most existing datasets focus on sports events, such as the Olympics, whereas datasets tailored for daily exercise activities remain scarce. Additionally, many of these datasets are unsuitable for direct application in AQA tasks. To address these limitations, we constructed a new AQA dataset, TaiChi-AQA, which includes detailed scoring annotations. Our dataset comprises 1313 Tai Chi action videos and features a comprehensive set of fine-grained labels, including action labels, action descriptions and frame-level perspective information. To validate the effectiveness of TaiChi-AQA, we systematically evaluated it using a variety of popular AQA methods. We also propose a straightforward yet effective module that integrates a multi-head attention mechanism with a gated multilayer perceptron (gMLP). This module is combined with the distributed autoencoder (DAE) framework. Extensive experiments demonstrate that our method achieves state-of-the-art performance on the TaiChi-AQA dataset. The dataset are publicly available at https://github.com/mlxger/TaiChi-AQA.

1 | Introduction

With the rapid development of computer vision and machine learning technologies, methods based on video analysis have been widely used in the field of sports science especially in posture recognition and action quality assessment. Computer vision techniques enable the identification and analysis of human movement patterns by processing image data in videos [1, 2].

As for action quality assessment (AQA), methods based on video analysis can quantitatively analyse athletes' movements to evaluate their quality and effectiveness. These machine learning approaches not only provide objective and consistent assessments but also serve as valuable tools to assist coaches in their work. For example, in various sports events [3–8], coaches can leverage the detailed analysis results generated by such methods to more accurately evaluate athletes' technical levels and training effectiveness, thereby enabling the design of more targeted training plans. Moreover, these approaches show significant potential in the field of healthcare [9, 10], providing strong support for sports rehabilitation and elderly health monitoring.

However, there are still challenges for AQA in multiapplication. This is mainly due to two reasons: (a) most existing AQA datasets are relatively limited in terms of video diversity and frame-level perspectives. (b) Some AQA video datasets [11, 12] may provide more complex scenes, but they only contain scores as the sole annotation, lacking detailed and diverse annotation information. To address these limitations, it is important to

Abbreviations: ANA, anti-nuclear antibodies; APC, antigen-presenting cells; IRF, interferon regulatory factor.

propose an AQA dataset that contains rich fine-grained information and more complex scenes.

Tai Chi, as an important part of traditional Chinese culture, is not only a form of martial arts but also a lifestyle that pursues physical and mental harmony. In recent years, with the growing emphasis on health and fitness, the popularity of Tai Chi has increased significantly, attracting more attention for its positive effects on physical and mental health [13–15].

However, the traditional Tai Chi teaching and training mainly rely on the verbal guidance and demonstration of coaches, which is subjective and makes it difficult to objectively assess and analyse the action quality. To solve this problem, it is important to introduce machine learning models for accurate evaluation and analysis.

Nevertheless, the primary challenge in building such an objective machine learning model is the deficiency of dataset. Therefore, we propose a TaiChi-AQA dataset, which contains short videos with rich annotated information and aims to support more accurate action quality assessment and analysis.

In the TaiChi-AQA dataset, each short video displays a 24-posture Tai Chi action with each video set in a unique scene, presenting a more complex background than most existing AQA datasets. The average video duration of this dataset is 14.45 s substantially longer than that of other AQA datasets. In addition, as shown in Table 1, the TaiChi-AQA covers detailed annotation information, including action type labels for each video and multiple fine-grained action scores. We also design the step-by-step descriptions for each action to describe the action of video presenter in detail, so as to provide a more detailed analysis of the presenter's actions. Overall, TaiChi-AQA stands out for its scene complexity and richer annotations compared to existing AQA datasets [3–5, 8, 11, 12, 16].

Although recent AQA methods [5, 18–22] have achieved encouraging results on relatively simple datasets by leveraging uncertainty modelling, contrastive learning and various attention mechanisms, they encounter two key limitations when migrating to complex scenarios such as TaiChi-AQA. First, prior studies [18, 19, 21] typically adopt global or single-scale spatiotemporal aggregation, which often ignores subtle level differences that are critical for Tai Chi quality assessment. Furthermore, directly using I3D (Inflated 3D ConvNet) features coupled with a multilayer perceptron (MLP) module dilutes the discriminative features and leads to poor performance when dealing with cluttered backgrounds and diverse motion patterns.

To address these limitations, particularly the insufficient accuracy of action recognition and the challenge of capturing key information in complex scenes, this study proposes a simple and effective multi-head attention mechanism-gated multilayer perceptron module (MHGM), consisting of a multi-head attention mechanism (MH) and a gated multilayer perceptron (gMLP) [23]. This design directly overcomes the identified gaps: the multi-head attention mechanism captures multiscale spatiotemporal dependencies and subtle level differences by attending to long-term relationships in action sequences, mitigating the issues of global or single-scale aggregation; meanwhile, the gMLP module refines the feature representations extracted from I3D, enhancing discriminative power in cluttered backgrounds and diverse motion patterns without diluting key features as in traditional MLP approaches. Specifically, we first utilise I3D technology to extract the feature vector of the action and then apply the multi-head attention mechanism to identify the long-term dependencies in the action sequence. Finally, we use the gMLP module to further refine the feature expression and enhance the model's ability to handle the complex actions.

Subsequently, a distributed autoencoder (DAE) [19] is used to map the learnt feature results to a Gaussian distribution. The quality of the Tai Chi movements practiced by the practitioner is evaluated by the score information in the Gaussian distribution. Our approach significantly enhances the model's performance in complex environments.

The contributions of this article can be summarised as follows:

- We introduce TaiChi-AQA, the first 24-posture Tai Chi video dataset with extensive fine-grained annotations specifically designed for complex scenes in action quality assessment. Compared to most existing datasets, TaiChi-AQA features not only a longer average video duration but also richer annotation information. Experimental results demonstrate its unique advantages.

TABLE 1 | Comparison between TaiChi-AQA and other sports-related AQA datasets.

Dataset	Duration	AVG DUR	Anno. type	Samples	Events[a]	Year
MTL dive [16]	15 m 54 s	6.0 s	Score	159	1	2014
UNLV dive [4]	23 m 26 s	3.8 s	Score	370	1	2017
AQA-7-dive [3]	37 m 31 s	4.1 s	Score	549	6	2019
MTL-AQA [4]	96 m 29 s	4.1 s	Action, score	1412	16	2019
Rhythmic gymnastics [12]	26 h 23 m 20 s	1 m 35 s	Score	1000	4	2020
FSD-10 [17]	—	3–30 s	Action, score	1484	—	2020
FineDiving [5]	3 h 30 m 0 s	4.2 s	Action, score	3000	30	2022
TaiChi-AQA (Ours)	**5 h 16 m 20 s**	**14.45 s**	**Action, score, step description**	**1313**	**24**	**2024**

[a]Definition of Events: In this table, 'Event' refers to the specific type of sport or action category represented in each dataset. Bold values indicate aspects unique to our TaChi-AQA dataset, highlighting differences from others (e.g., MTL dive only includes diving, whereas our TaChi-AQA includes 24 diverse categories).

- We propose a simple and efficient multi-head attention mechanism-gated multilayer perceptron module (MHGM), which significantly enhances feature learning accuracy and action quality assessment in complex scenes.
- Our experimental results reveal that, compared to existing action quality assessment methods, the proposed model achieves substantial performance improvements on the TaiChi-AQA dataset reaching a state-of-the-art level in the field.

2 | Related Theories

2.1 | Sport Video Dataset

Action recognition in professional sports video is a research hotspot in the field of computer vision. Compared to action understanding in general video datasets, this task is more challenging. Especially, action understanding in competitive sports videos heavily relies on comprehensive and diverse sports datasets. In early research, Niebles et al. [24] introduced the Olympic sports dataset into sports modelling. Subsequently, Karpathy et al. [25] constructed a large Sports 1M dataset and achieved great improvement on performance through a powerful feature extraction method. For the AQA task, there are also many datasets related to sports and exercise, such as the multisports scene dataset (AQA-7) proposed by Parmar and Morris [3] and another larger multitask dataset (MTL-AQA) [4]. Xu et al. [5] first proposed a large-scale diving action dataset (FineDiving), whereas Liu [17] proposed a figure skating action dataset FSD-10 that includes action types and scores. Shao et al. [26] proposed the FineGym dataset, which provided the coarse-to-fine temporal and semantic annotations for action recognition. Zhang et al. [27] proposed a multiperson long video dataset LOGO that was used to analyse the relationship between group athletes.

Nevertheless, most of the above datasets are limited to single-perspective AQA tasks in sports events where action quality assessment is conducted using video data captured from only one camera angle or viewpoint rather than utilising multiview or multiperspective data. We developed TaiChi-AQA, a video dataset specifically designed for Tai Chi exercise. This dataset includes fine-grained action category labels, action step description labels, and scoring labels that can be directly used for action quality assessment. In Table 1, the differences between TaiChi-AQA and other sports-related AQA datasets are presented in detail. Here, the term 'Event' is defined as the specific sport type or action category.

2.2 | Action Quality Assessment

Action Quality Assessment (AQA) has emerged as a prominent research area in computer vision aiming to quantify the execution quality of actions in videos. Recent advancements in deep learning have driven significant progress with studies focussing on diverse tasks such as sports analysis [18, 19] and skill evaluation [28, 29].

In early research, Pirsiavash et al. [16] first proposed the concept of AQA and mapped the pose-based features and hierarchical convolutional features to SVR scores. Since then, AQA has attracted much attention and developed rapidly. The mainstream method describes it as a direct modelling regression between action features and referee scores. For instance, Parmar and Tran Morris [8] first made use of deep learning models and proposed C3D-SVR and C3D-LSTM to predict scores.

To mitigate ambiguities arising from subjective referee evaluations, several approaches have shifted towards probabilistic modelling. Tang et al. [18] introduced uncertainty-aware score distribution learning (USDL), framing AQA as learning a probability distribution over possible scores. They extended this to Multi-USDL (MUSDL) to simulate multireferee scoring in real-world scenarios. Similarly, Zhang et al. [19] proposed a distribution autoencoder (DAE) that encodes videos into score distributions and employs re-parameterisation for sampling, enabling more accurate video-score mappings.

Beyond direct regression, contrastive frameworks learn relative quality by comparing video pairs. For instance, Yu et al. [20] proposed a new contrastive regression framework (CoRe) to learn and predict the differential scores between paired input videos. Ke et al. [30] proposed a T^2CR network that fuses direct and contrastive regression across multiple visual fields to enhance action quality assessment. Liu et al. [31] proposed a Replay-Guided Triple-Stream Contrastive Transformer augmented with a Temporal Concentration Module, leveraging multiview replay videos and concentrated error/highlight features. Furthermore, they designed a Hierarchical Joint Training Replay-guided Contrastive Transformer [32] that leverages slow-motion replays as privileged self-supervision.

Recent works have emphasised semantic fine-grained analysis to capture subtle action differences. Xu et al. [5] introduced the FineDiving dataset, which leverages fine-grained action decomposition to detect step-level variations in paired videos. Zhou et al. [21] proposed a Hierarchical Graph Convolutional Network (HGCN) that aggregates sequential action features across scenes, predicts score distributions and enhances discriminative representations for improved performance. Extending graph-based methods, Yun et al. [33] presented a teacher-reference-student architecture that leverages both labelled and unlabelled data using teacher and reference networks to generate pseudo-labels and a confidence memory to enhance their reliability for semi-supervised action quality assessment.

The advent of Transformer architectures has further advanced AQA by enabling long-range dependency modelling. Wang et al. [34] proposed TSA-Net, a Tube Self-Attention Network that embeds a single-object tracker and a Tube Self-Attention module to exploit sparse spatiotemporal interactions. Bai et al. [22] introduced the temporal parsing transformer, which parses action sequences temporally to assess quality, effectively handling long-range dependencies in complex actions. Most recently, Xu et al. [35] proposed FineParser, a finegrained spatio-temporal action parser that employs dual transformer decoders to extract aligned human-centric foreground representations, suppressing background interference and yielding precise action quality scores.

In parallel, action quality assessment using AI-generated videos and multimodal approaches has enriched AQA methodologies. Chen et al. [36] proposed GAIA, a large-scale causal-reasoning-based dataset comprising 971k human ratings on AI-generated action videos, which exposes the failure of existing AQA and T2V metrics and establishes a new benchmark for evaluating action quality in AI-generated video. Concurrently, Zhang et al. [37] proposed Narrative Action Evaluation (NAE), a prompt-guided multimodal framework that assesses action quality in storytelling contexts. Although these works lie outside traditional AQA, they highlight the need for robust, generalisable models.

Building upon these methods, this paper adopts techniques inspired by USDL, DAE and HGCN [18, 19, 21], incorporating the MHGM (Multihead-attention-Gated MLP) module to model and perform action quality assessment in complex scenes.

3 | Tai Chi Dataset

3.1 | Dataset Construction

This study constructs a novel 24-posture short video Tai Chi dataset and explains the Tai Chi movements and their scoring in detail. This dataset constitutes a challenging benchmark for action quality assessment (AQA). In this section, we will discuss the dataset in detail, focussing on data collection, scoring criteria and label annotations.

3.1.1 | Dataset Collection

As shown in Figure 1, the TaiChi-AQA covers 24 different Tai Chi movements with a total of 55 long videos, including Starting Posture, left and right Wild Horse's Mane, White Crane Spreads Its Wings and more. These videos capture movements from different perspectives, including the front, left side, right side and the back.

The video materials in the dataset are derived from video materials on platforms such as YouTube and Bilibili. The majority of these videos were collected from publicly available sources uploaded by different individuals and organisations. As a result, we do not have access to detailed information about the specific camera models or equipment used during recording. Nevertheless, our dataset comprises videos with high resolution (such as 720p and 1080p) and frame rates (such as 25 fps and 30 fps). The features extracted from these videos using the I3D model will be made publicly available for further research. For action analysis, we initially sample the video at a sampling rate of 16 fps, then split the long video into 1313 short videos according to the different actions and further perform fine-grained action annotation and labelling construction at a sampling rate of 1 fps.

Each video contains action annotation, action category, action description, module score of exercise level, action step quality module score, final score under two different weights and multiperspective frame information. The detailed annotations and scoring information of the TaiChi-AQA dataset is defined as follows:

- *Action Annotation*: A unique identifier assigned to each video for easy distinction and retrieval.
- *Action Category*: The specific Tai Chi movement category to which the video belongs.
- *Step Description*: A brief textual description of the step of action performed in the video
- *Exercise Level*: Evaluation of the performer's exercise level based on the following criteria:
  - — *Body Posture*: Accuracy and standardisation of body posture.
  - — *Mental Outlook*: Display of confidence and focus during the performance.
  - — *Strength Control*: The ability to control strength for maintaining balance and fluidity.
  - — *Coordination*: Coordination of body movements during the action.

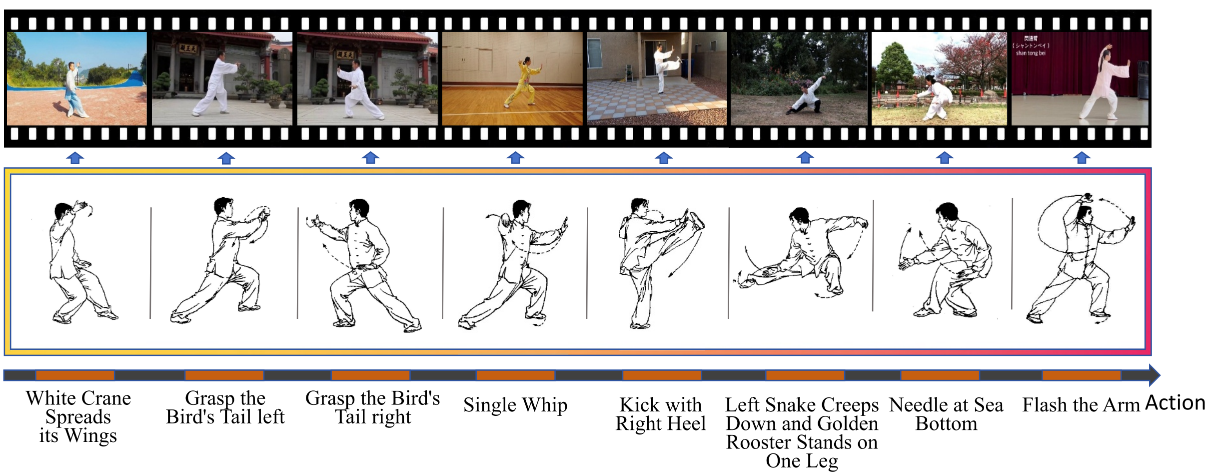

FIGURE 1 | Overview of the Tai Chi-AQA dataset: The figure depicts 8 representative movements, White Crane Spreads Its Wings, Grasp the Bird's Tail: (left and right), Single Whip, Kick with Right Heel, Left Snake Creeps Down and Golden Rooster Stands on One Leg, Needle at Sea Bottom and Flash the Arm Action.

- *Action Quality*: Assessment of the quality of actions using the following dimensions:
  - *Movement Fluency*: The continuity and fluency of movements.
  - *Balance*: Stability when performing the action.
  - *Shape and Skill*: The degree of adherence to gesture standards.
  - *Step Patterns and Footwork*: The degree of compliance with the standard footwork.
  - *Body Shape and Posture*: Overall aesthetic appeal and conformity to posture standards.
- *Final Score*: The overall evaluation of the performance under two different weighting standards:
  - *Score 1*: final score under the first weighting standard.
  - *Score 2*: final score under the second weighting standard.
- *Visual Perspective*: multiperspective visual information captured in the videos.
- *Video Duration*: total length of the video (seconds).

3.1.2 | Formulation of Scoring Rules

According to the current 'Martial Arts Competition Rule' [38, 39] in China and the research literature on Tai Chi movement evaluation indicators [40], this study synthesised the fundamental requirements of Tai Chi training. Furthermore, three sport experts formulated the scoring rules for Tai Chi movements. The evaluation of 24-posture Tai Chi includes basic boxing posture, footwork, support, balance and other key elements.

During the execution, it is required that the basic movements of the trainees, such as hand shape, step pattern, footwork, body shape and posture, must meet the established standards. At the same time, trainees are required to maintain correct posture, control their strength, coordinate their body and display a positive mental outlook during practice. Based on the above requirements, the study develops two scoring dimensions for Tai Chi: Action Quality (AQ) and Practice Level (PL).

The accuracy of movements (e.g., whether the movements meet their requirements, follow the correct sequence and avoid incoherence) forms the basis of the AQ score. On the other hand, PL is assessed based on specific criteria such as fluency, strength control, rhythm and overall performance. Table 2 shows the scoring detail of the Tai Chi Action Quality, and Table 3 presents the scoring detail of the Tai Chi Practice Level.

These sublabels are not introduced by the authors but are standard in Tai Chi evaluation. Specifically, the four sublabels for PL (body posture, mental outlook, strength control and coordination) follow the official scoring rubric of the Chinese Wushu Association Tai Chi Competition Rules (2018) [41], which are widely adopted in sanctioned Tai Chi tournaments. Similarly, the five sublabels for AQ (movement fluency, balance, hand shapes, step forms and body shapes) are derived from established Tai Chi standards in the 'Martial Arts Competition Rule' in China and related research literature [42, 43].

Based on the scoring rules in Tables 2 and 3, we adopt 4 sublabels under the Practice Level module to evaluate the trainee's

TABLE 2 | Scoring details of Tai Chi action quality.

Items	Rules	Deduction value
Body shape and posture	The body posture should be proper, with good posture of the head, trunk and limbs, and there should be no obvious looseness or collapse in the movement. The body should maintain a stable and proper posture.	0.1 points deducted for each instance of nonconformity.
Step patterns and footwork	The footwork should be natural, and the step patterns should conform to the Tai Chi standard. The steps should be light, coherent and precise, and the transition between steps should be smooth without jumping or imbalance.	0.1 points deducted for each instance of nonconformity.
Hand shape and skill	The hand shape should be up to the specification, and the movements should be in place. The palm and the fingers should be in correct position, and the strength and speed should be mastered. The hand movements should be consistent without any shaking and dislocation.	0.1 points deducted for each instance of nonconformity.
Balance	A good balance should be kept during the movement, and the centre of gravity should not shift or fall. Whether it is in static or dynamic movements, the body should maintain stable especially in the movements of turning or knee raising.	0.1 points deducted for each instance of nonconformity.
Movement fluency	The transition between movements should be natural and smooth, and the whole progress should be coherent, avoiding obvious pauses or stiffness. The rhythm of the movements should be even without any rapid or slow phenomena.	0.1 points deducted for each instance of nonconformity.

TABLE 3 | Scoring detail of the Tai Chi practice level (aligned with CWA 2018 [41] and IWUF 2018 [42]; total score 0–20 via summation of four dimensions: body posture, mental outlook, strength control, coordination, each 0–5).

Grade	Level	Score range	Scoring standard
Good	Level 1	4.21–5.00	Excellent (minimal errors, < 5% deviation): Body posture—precise with no deviation; mental outlook—strong spirit (focused gaze, fluid continuity); strength control—consistent power (no trembling > 5s holds); coordination—full sync (rhythm error < 2%).
	Level 2	3.01–4.20	Good (minor errors, 5%–10% deviation): Body posture—accurate with slight deviation; mental outlook—clear focus and spirit; strength control—steady power (minimal trembling 3–5s holds); coordination—high sync (rhythm error < 5%).
	Level 3	2.51–3.00	Sufficient (acceptable errors, 10%–15% deviation): Body posture—basically in place; mental outlook—moderate spirit expression; strength control—adequate power (stable 2–3s holds); coordination—basic sync (rhythm error < 10%).
Average	Level 4	2.31–2.50	Moderate (some errors, 15%–20% deviation): Body posture—acceptable with deviations; mental outlook—average spirit (intermittent focus); strength control—sufficient but occasional weakness (1–2s holds); coordination—acceptable sync (rhythm error 10%–15%).
	Level 5	2.11–2.30	Average (frequent minor errors, 20%–25% deviation): Body posture—basically in place but variable; mental outlook—basic spirit; strength control—variable power (occasional instability); coordination—partial sync (rhythm error 15%–20%).
	Level 6	1.91–2.10	Below-average (noticeable errors, 25%–30% deviation): Body posture—often not in place; mental outlook—weak spirit (focus lapses); strength control—inconsistent power (frequent trembling); coordination—limited sync (rhythm error 20%–25%).
Not good	Level 7	1.71–1.90	Low (significant errors, 30%–35% deviation): Body posture—inaccurate; mental outlook—minimal spirit; strength control—insufficient power (< 1s unstable holds); coordination—poor sync (rhythm error 25%–30%).
	Level 8	1.51–1.70	Very low (major errors, 35%–40% deviation): Body posture—highly inaccurate; mental outlook—poor spirit; strength control—weak with interruptions; coordination—disrupted sync (rhythm error > 30%).
	Level 9	0–1.50	Inadequate (extensive errors, > 40% deviation): Body posture—extensive deviations; mental outlook—absent spirit (disjointed); strength control—minimal stability; coordination—no effective sync.

exercise quality and 5 sublabels under the Action Quality module to assess the execution quality of movements. The final score is calculated using all 9 sublabels as illustrated in Figure 2.

In order to ensure the rigour and repeatability of the scoring method, we provide a detailed explanation of the scoring process. As detailed in Table 3 and illustrated in the left half of Figure 2 (which shows both per-sub-label and total module ranges), the score of the Practice Level (PL) module is obtained by adding the scores of the four submodules: body posture (bp), mental outlook (mo), strength control (sc) and coordination (co) as shown in Equation (1). The score range of each of these four submodules is 0–5 points, resulting in a total score range for the Practice Level module of 0–20 points.

$$PL = bp + mo + sc + co \tag{1}$$

As detailed in Table 2 and illustrated in the right half of Figure 2, the score of the Action Quality (AQ) module is obtained by adding the scores of the five submodules: movement fluency (mf), balance (ba), hand shapes (hs), step forms (sf) and body shapes (bs) as shown in the following Equation (2). The score range of these five submodules is 0–1 points, and thus the total score range of the action quality module is 0–5 points.

$$AQ = mf + ba + hs + sf + bs \tag{2}$$

The total score range of the Practice Level (PL) module is 0–20 points, and the total score range of the Action Quality (AQ) module is 0–5 points. According to the research literature on the evaluation indicators of Tai Chi [40], the final score is obtained by directly adding the scores of the Practice Level (PL) module and the Action Quality (AQ) module with the final score range of 0–25 points. However, in the AQA task, greater emphasis is required on the level of Action Quality. Therefore, we need to adjust the score range of the Action Quality module from 0 to 5 points to 0–20 points through multiplying by a coefficient w_2 so as to increase the proportion of Action Quality in the final score.

$$FinalScore = w_1 * PL + w_2 * AQ \tag{3}$$

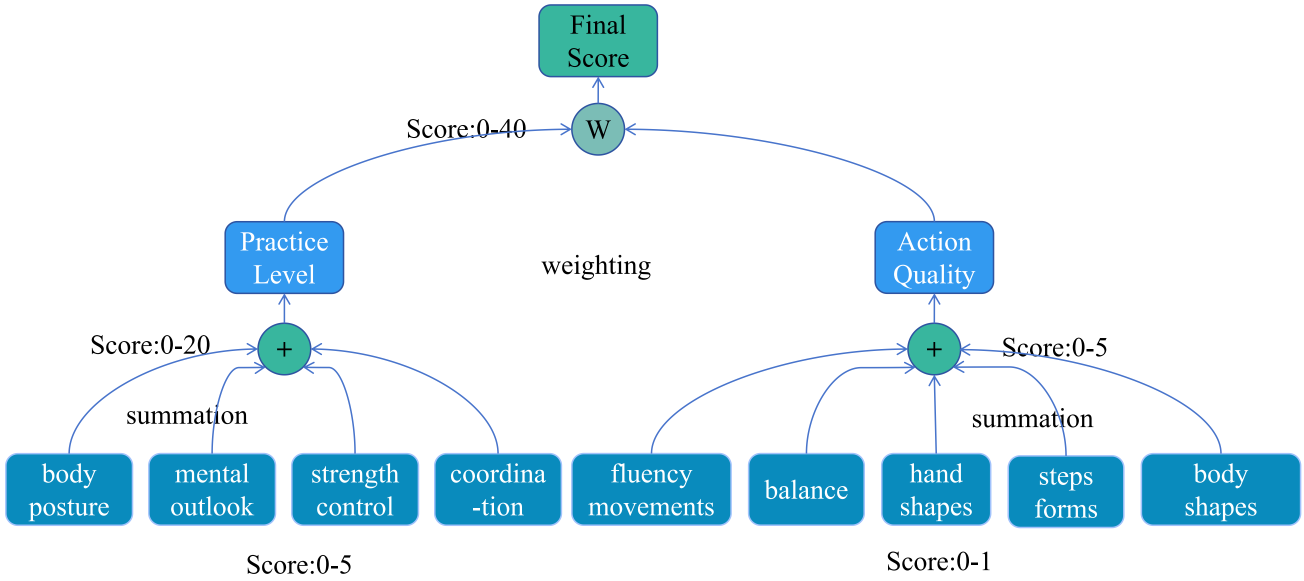

FIGURE 2 | Hierarchical structure for calculating the final score in Tai Chi performance evaluation. The final score is computed as the weighted sum of the Practice Level (PL) and Action Quality (AQ) modules. PL (range: 0–20 points) is the summation of four sub-modules: body posture (bp), mental outlook (mo), strength control (sc) and coordination (co), each scored 0–5 points. AQ (range: 0–5 points) is the summation of five submodules: movement fluency (mf), balance (ba), hand shapes (hs), step forms (sf) and body shapes (bs), each scored 0–1 point. Weighting factors are set to $w_1 = 1$ and $w_2 = 4$ to emphasise AQ resulting in a final score range of 0–40 points.

In the above Equation (3), Final Score represents the final score, whereas w_1 and w_2 are the coefficients used to adjust the proportions of PL and AQ in the final score. To ensure a 1:1 ratio between PL and AQ, we set the value of w_1 to 1 and w_2 to 4. With these adjustments, the final score range is calculated to be 0–40 points.

3.1.3 | Label Annotation

This study focuses on the richness and diversity of Tai Chi movements using manual annotation to label the movements in the video dataset accurately. Based on the categorisation of Tai Chi movements, 55 long videos are segmented and disassembled into 1313 short video clips. To ensure reliable scoring, the assessment of Tai Chi was conducted by one judge with national-level Tai Chi professional qualifications and three judges with relevant professional qualifications. Each expert judges according to Tai Chi standards and is reviewed by another expert. For privacy reasons, the judges' specific personal information cannot be disclosed.

According to Table 2, five fine-grained scoring indicators are defined for Action Quality: body shapes, step forms, hand shapes, balance and movement fluency. Similarly, Table 3 defines four fine-grained scoring indicators for Practice Level: body posture, strength control, coordination and mental outlook. Based on the labelling results of the nine labels, we calculate the two module scores (AQ and PL) and the final score of the nine labels, which can be the freely available action quality assessment labels.

To ensure the accuracy, the labelling results provided by one expert were reviewed and adjusted by another expert achieving a double-review process. Furthermore, in order to enhance the accessibility of Tai Chi movements, an action step description label is provided for each short video. We make a detailed description for each step to enrich the fine grain of the dataset as shown in Figure 3.

Finally, this study constructs a high-quality short video dataset, that is, TaiChi-AQA, the Tai Chi dataset with multiple labels and its open-source website is: https://github.com/mlxger/TaiChi-AQA.

3.2 | Dataset Statistics

The dataset constructed in this study covers 24 different action categories and contains 1313 video samples. The average duration of each video is 14.15 s, the dataset comprises 17 annotations, 24 action types and four different perspectives as illustrated in Figure 2.

Figure 4 shows the distribution of the final score ranging from 0 to 40 points. It can be seen in the video score distribution that the overall distribution is between 25 and 40 points, and 33 and 39 points are the two main peaks with the largest number of videos. There are fewer samples in the low score area, whereas the majority are in the medium-to-high score range.

The Tai Chi dataset, TaiChi-AQA, differs significantly from the existing AQA dataset in terms of annotation types and action diversity. Specifically, it can be explained from the following aspects:

- *Multi-dimensional scoring labels*: The dataset consists of short videos annotated with multiple scoring labels. These scoring labels can be used in combination under different scoring rules, which helps to diversify the assessment. In contrast, other fine-grained sports datasets are difficult to

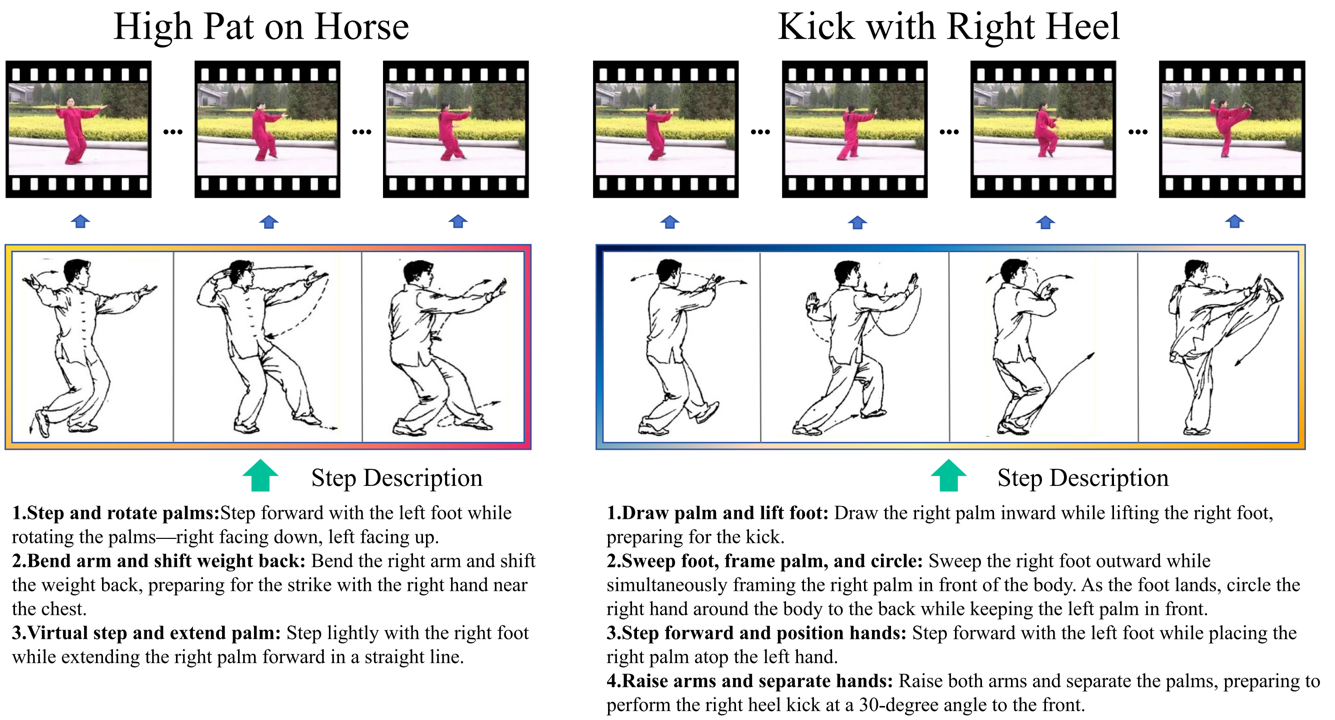

FIGURE 3 | The left picture is the Step Description of High Pat on Horse containing three steps, and the right picture is the Step Description of Kick with Right Heel containing four steps.

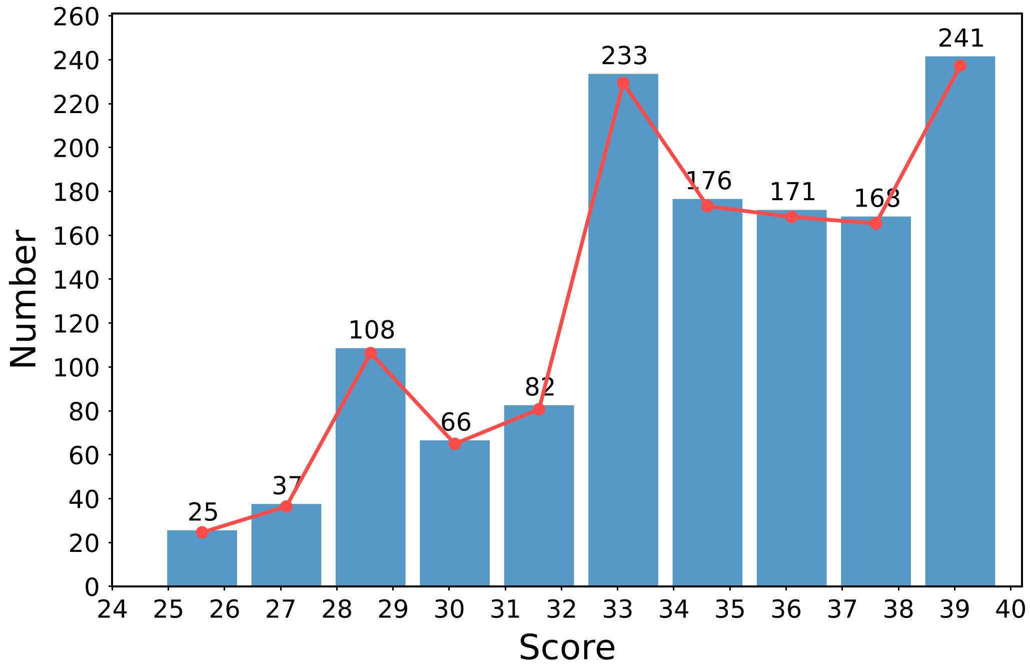

FIGURE 4 | The score distribution of videos. The horizontal axis (X axis) is the score of the video, and the vertical axis (Y axis) is the number of samples of the video.

be directly used for action quality assessment due to the lack of multi-dimensional scoring labels.

- *Action categorisation and identification*: The dataset includes both action identification and category labels, allowing for clear differentiation between action types and attributes. In addition, the dataset also contains step descriptions, which can be used to study the fine-grained feature extraction of actions and to analyse the correlation between action steps.
- *A novel fine-grained sports video dataset*: The TaiChi-AQA dataset is the first fine-grained sports video dataset specifically designed to evaluate Tai Chi movements. It can

effectively fill the gap of fine-grained group annotation in the AQA field.

3.3 | Dataset Assessment

The image dataset constructed in this study adopts a frame sampling rate of 1 fps. To ensure the quality of the extracted frames, we employ the structural similarity index measure (SSIM) [44] to quantitatively assess the visual similarity between the frames extracted after video segmentation and the corresponding original video frames. This evaluation verifies that the segmentation process preserves the structural and perceptual integrity of the images, minimising any degradation or artefacts introduced during frame extraction. Additionally, by comparing SSIM [44] values of successive extracted frames, we assess the stability and consistency of the segmentation results, evaluating robustness in handling complex scenarios such as moving objects, changing backgrounds and varying lighting conditions.

SSIM [44] comprehensively involves the brightness, contrast and structure information of the image, aiming at more accurately reflecting the perception of the human eyes. The calculation formula of SSIM is as follows:

TABLE 4 | Meaning of *SSIM* value.

Range of *SSIM* value	Meaning
0.0~0.50	Poor segmentation effect
0.51~0.70	General segmentation effect
0.71~0.84	Good segmentation effect
0.85~1.00	Excellent segmentation effect

$$SSIM(X, Y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4)$$

In the above Equation (4), X and Y are two images to be compared. u_x and u_y are, respectively, the average brightness of the two images. σ_x and σ_y are the brightness variances of images X and Y. σ_{xy} is the covariance of images X and Y. C_1 and C_2 are constants introduced to avoid the denominator being zero, which are usually the small constants.

The range of SSIM [44] values is [0, 1], which can be divided into four groups to represent different levels of segmentation effects as shown in Table 4. In this study, each short video is segmented into a picture set, and two pictures in each picture set are selected as the input of the formula SSIM. In the end, the average SSIM value of the entire picture set is calculated.

For the image sets corresponding to the 24 actions, the average SSIM value reaches 0.85427 falling within the [0.85, 1) range. This indicates that the segmentation of the Tai Chi dataset is effective, preserving high visual fidelity and making it suitable for AQA tasks.

4 | Scheme of This Study

In this section, we will systematically introduce our method. The overall architecture of our method is shown in Figure 5.

First, the video is divided into multiple clips, each of which is used as an independent input. Next, I3D (Inflated 3D ConvNet) is applied to each clip to extract spatiotemporal features, which captures the dynamic information in the video by expanding the 2D convolution to 3D convolution. After, these features assign different weights to the features of each clip through a multi-head

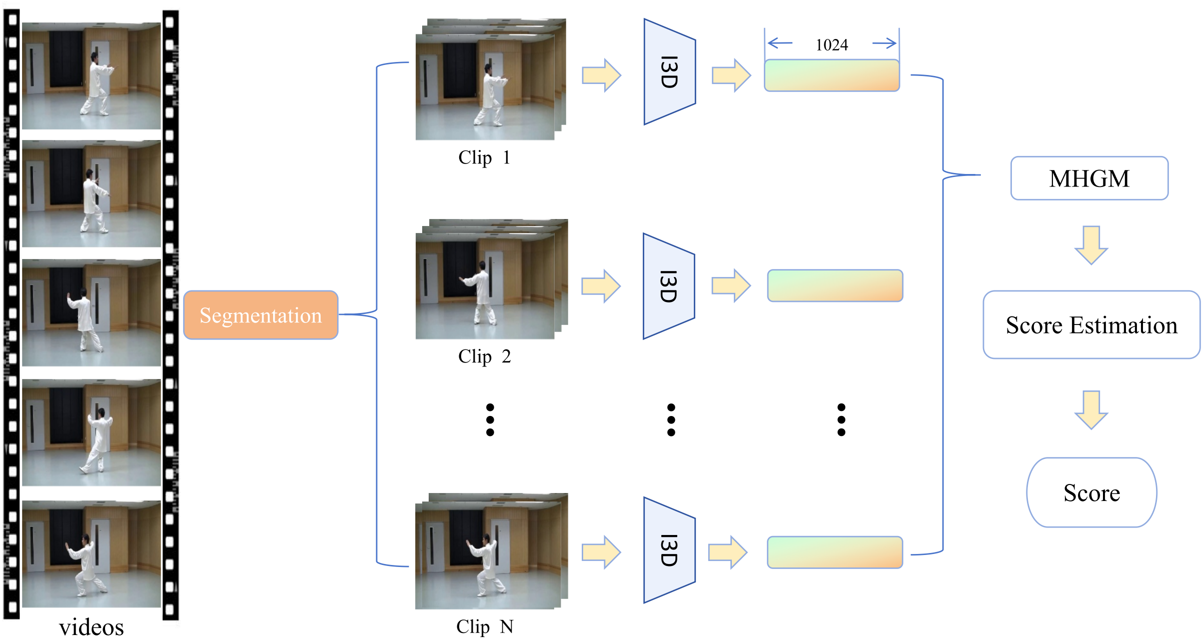

FIGURE 5 | Overall network architecture.

attention mechanism, thereby focussing on the information that contributes most to the score. Subsequently, the stacked gated multilayer perceptron (gMLP) performs further nonlinear processing on the attention-weighted features, emphasising key features and suppressing irrelevant features to improve the expressiveness of the features. Finally, the extracted features are mapped to a Gaussian score distribution through a score distribution regression autoencoder (DAE), and the Final Score of the video is obtained by sampling from the score distribution.

We record the video frame index of each frame and create the required fine-grained pickle serialisation file of the dataset according to the experimental requirements, including 19 annotations. In order to ensure that the ratio of training set and test set is 3:1 in the experiment, we randomly shuffle the 1313 video frame sets, selecting the first 985 ones as the training set and the remaining 328 as the test set. Moreover, the shuffled results are also saved as pickle serialisation files, so that the experimental results are easy to read and compare.

4.1 | Data Preprocessing

In TaiChi-AQA dataset, different action categories are labelled as shown in Table A1. Frames are selected from the video time series at fixed intervals using a technique called equidistant frame extraction. This method is very effective in processing video sequence data, which ensures that the selected frames are representative while avoiding data redundancy. The formula for equidistant frame extraction is as follows:

$$F_i = \left\lfloor i \times \frac{N}{M} \right\rfloor \tag{5}$$

In the above Equation (5), F_i is the index of the ith frame of the video, N is the total number of video frames and M is the number of frames to be extracted. For this study, M is set to 103 based on experimental requirements. Every 16 frames are extracted as a segment, each segment contains the latter half of the previous segment, and the last segment contains the last 16 frames of the selected video sequence.

4.2 | Feature Extraction and MHGM Module

I3D is a deep learning model designed for video understanding tasks, possessing strong spatiotemporal feature learning capabilities. The I3D model extracts joint spatial topological relations and short-term motion patterns from continuous video clips, ultimately outputting 1024-dimensional temporal features from its final convolutional layer (Conv5).

Each different clips of Tai Chi action videos is processed by the I3D model to extract the corresponding spatiotemporal features, and all segments share the precise weights of the fully connected layer. Once we obtain the feature vectors of all video segments, we concatenate them together. Next, a multi-head attention mechanism is applied to assign different weights to the features of each clip, focussing on the information most relevant to the score. Then, the average of the features of all clips is taken as the final feature vector of the action video to ensure that the information of each video clip is equally considered. The pipline of inputting spatiotemporal features into the MHGM module and the architecture of MHGM are shown in Figure 6.

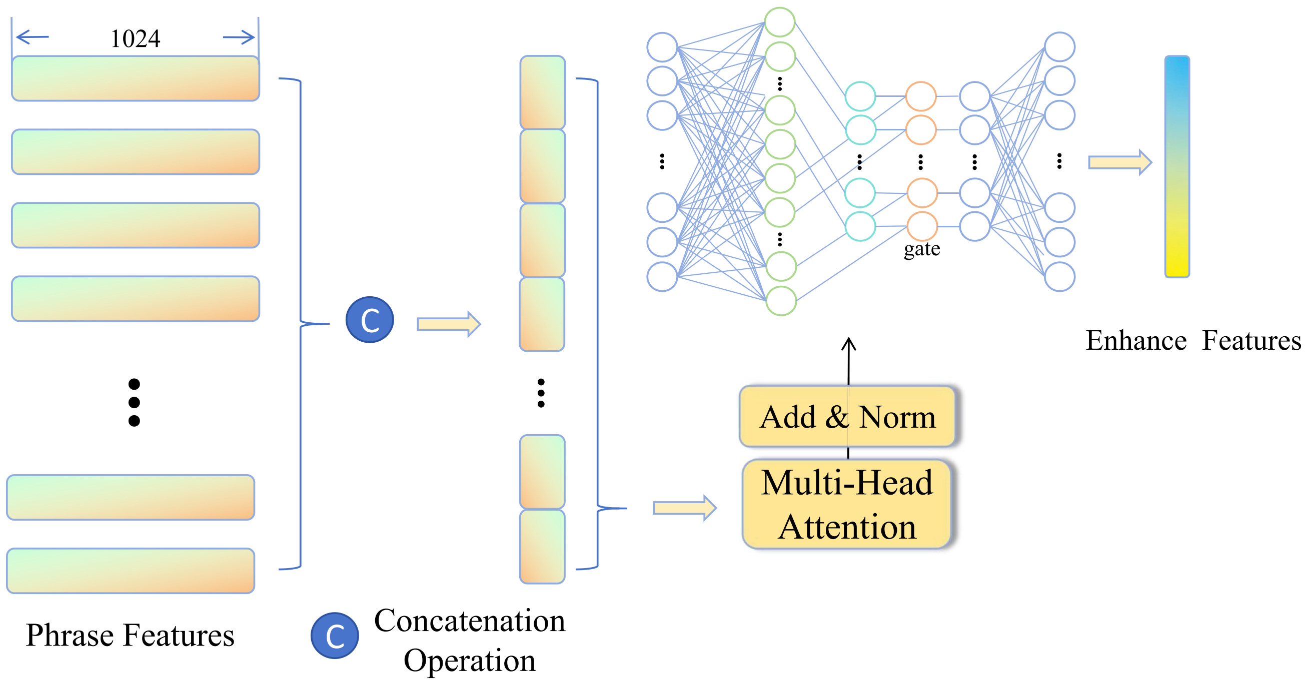

FIGURE 6 | The pipline of inputting spatiotemporal features into the MHGM module.

In order to allow the model to capture features from different subspaces, the multi-head attention mechanism (MH) utilises the self-attention mechanism (SA) independently for each head and then combines the outputs of all heads. The formula for the multi-head attention mechanism is as follows:

$$\mathrm{MH}(Q, K, V) = [\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_h] \cdot W^O \tag{6}$$

Among them, W^O is the weight matrix used to connect the results of each head, and each head_i is an independent self-attention mechanism (SA). The formula of self-attention mechanism is as follows:

$$\mathrm{SA}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) \cdot V \tag{7}$$

In the above Equation (7), as the m-dimensional feature is $[x_1, x_2, \ldots, x_n]$, each element x_i is first projected onto three different vectors: query vector Q, key vector K and value vector V. $\sqrt{d_k}$ is a scaling factor used to avoid excessive similarity values. The softmax function converts the similarity values into weights, which are then multiplied by V to obtain a weighted value vector.

$$\mathrm{head}_i = \mathrm{SA}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{8}$$

In the above Equation (8), W_q, W_k and W_v are the learnable weight matrices with each head head_i using an independent self-attention mechanism (SA).

After passing through the multi-head attention mechanism module, the 1024-dimensional features should also get through a multi-layer perceptron (MLP) with a gated unit. The gated multilayer perceptron (gMLP) allows the model to dynamically adjust the importance of features when processing the input features. Through the gating signal, the model can selectively enhance or suppress certain features. This mechanism can improve the model's adaptability to input data, thereby improving the performance when processing score prediction tasks.

Specifically, the input m-dimensional features are expanded through the linear layer, and then the extended output is split into two parts by the chunk method: gate and content. Each part contains half of the expanded features.

$$\mathrm{output} = \mathrm{gate} * \mathrm{content} \tag{9}$$

In the above Equation (9), gate is the gated signal calculated by the input features. Content is the original input feature or the transformed feature.

The activation function *GELU* is applied to gate, and then the result is multiplied by content element by element. Such a gated mechanism allows the model to perform nonlinear transformation and selective amplification on the features before passing the second linear layer. Finally, the enhance feature dimension is compressed back to the original m-dimensional output through a second linear layer. This ensures that the model focuses on features prioritised by the multi-head attention mechanism.

4.3 | Score Prediction and Loss Function

Our goal is to predict the quality score of action videos, but this score is inherently uncertain. This uncertainty may come from the noise of the data itself, such as differences in subjective judgements. As a result, simply predicting a fixed score is insufficient. Instead, the ideal approach is to predict the probability distribution of the score, providing additional information about prediction uncertainty.

After learning the features through a multi-head attention mechanism and a gated multilayer perceptron (gMLP) model, we introduce a score distributed autoencoder (DAE) to map the extracted features into a Gaussian score distribution to better capture this uncertainty. The score prediction architecture is shown in Figure 7.

The input 1024-dimensional feature vector is encoded as distribution parameters $\mu(x)$ (mean) and $\sigma^2(x)$ (variance) through a

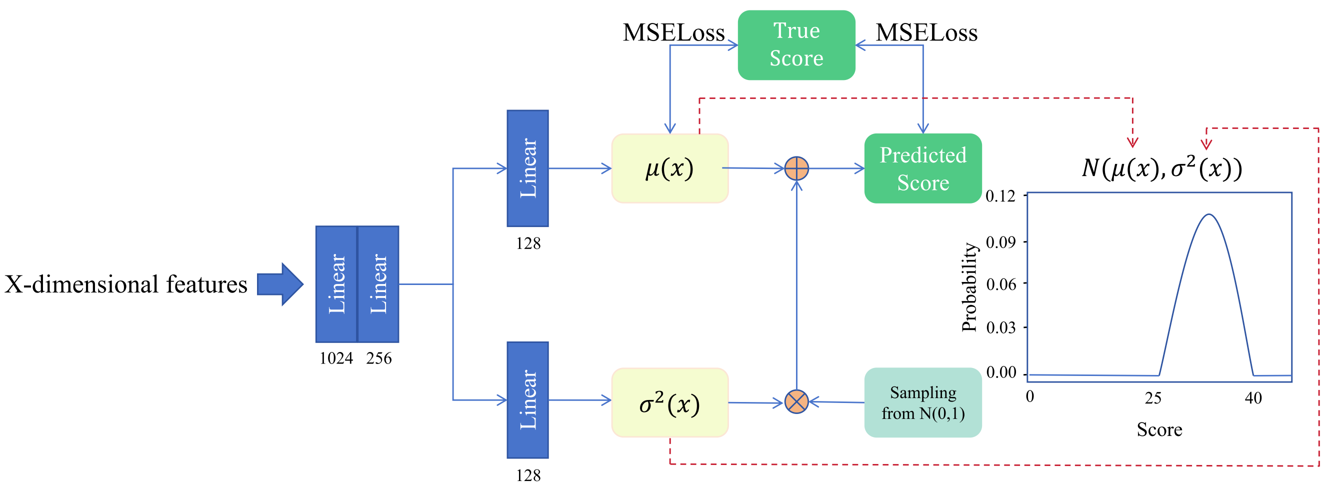

FIGURE 7 | Architecture diagram for score prediction.

neural network, where the mean $\mu(x)$ represents the predicted score and the variance $\sigma^2(x)$ represents the uncertainty or confidence level of the prediction. The 1024-dimensional video feature x is encoded into a random variable y through a probability encoder $g(y;\theta(x))$.

$$g(y;\theta(x)) = \frac{1}{\sqrt{2\pi\sigma^2(x)}}\exp\left(-\frac{(y-\mu(x))^2}{2\sigma^2(x)}\right) \tag{10}$$

Among them, $\mu(x)$ and $\sigma^2(x)$ are the mean and the variance of the Gaussian distribution both determined by the feature vectors x.

To generate samples from a Gaussian distribution as the prediction scores, the DAE first draws a random sample from the standard normal distribution $N(0,1)$. Then, y is calculated according to the sampled random variable, mean parameter $\mu(x)$ and variance parameter $\sigma^2(x)$ using the reparameterisation trick [45] employed by the DAE as shown in the following formula:

$$y = \mu(x) + \varepsilon \cdot \sigma(x) \tag{11}$$

where ε is the sampled random variable from $N(0,1)$.

In order to ensure that the predicted distribution is as close as possible to the distribution of the actual data, and to incorporate the reconstruction error in the prediction model to minimise the difference between the input data and the reconstructed data, we calculate the mean square error between the predicted score and the actual score true. It ensures that the model can effectively reconstruct the input data and add the mean square error between the actual score true and the mean $\mu(x)$, so that the model's predicted mean is consistent with the input data, thereby improving the stability and accuracy of the model.

Our loss function consists of these two losses, defined as follows, where N is the number of batches:

$$\text{Loss} = \frac{1}{N}\sum_{i=1}^{N}\left[(\text{pred}-\text{true})^2 + (\text{true}-\mu(x))^2\right] \tag{12}$$

5 | Experiment

5.1 | Data and Experimental Environment

The dataset used in this paper is the TaiChi-AQA dataset with fine-grained annotations. The experimental environment is an Intel(R) Xeon(R) Silver 4310 CPU @ 2.10 GHz processor with 26G memory. The GPU is RTX 4090(24 GB) * 1, PyTorch is 1.13.1 and Python is 3.8(ubuntu22.04).

Table 5 describes the dataset composition of each action in detail, covering the training set and test set of the action. Each action category corresponds to a specific movement in the 24-posture Tai Chi. The training set and test set are saved by randomly shuffling the entire dataset with a ratio of approximately 3:1.

5.2 | Evaluation Metrics

We comprehensively evaluate our AQA method under multiple metrics, including Spearman's rank correlation coefficient (ρ), coefficient of determination (R^2), mean absolute error (MAE), root mean square error (RMSE) and relative L2 distance (R-ℓ_2 scaled by $\times 10^2$).

TABLE 5 | The number of individual movements in the training set and test set is provided (refer to the complete list of movement abbreviations and their corresponding full names in Table A1).

Action category	Number of training set	Number of test set
CF1	47	8
PWM	39	16
WCSW	42	13
BKTS	45	10
PP	44	11
RM	41	13
GBT-L	39	16
GBT-R	39	16
SW1	45	10
WHLC	45	10
SW2	43	12
HPH	42	13
KRH	34	21
SOEBF	37	18
TKLH	41	14
LSCD	41	14
RSCD	37	13
SBF	43	12
NSB	39	15
FTB	40	14
TDPP	42	12
AC	41	13
CH	40	14
CF2	39	15

Spearman's rank correlation coefficient (ρ) is used to evaluate the rank correlation between the predicted value and the actual value, whereas the coefficient of determination (R^2) measures the model's ability to explain the variance in the true values.

In the prediction score regression task, we use MAE and RMSE to quantify the gap between the predicted score and the true score. MAE captures the average absolute discrepancy, whereas RMSE emphasises larger errors due to its squared nature. Additionally, the relative L2 distance (R-ℓ_2) provides a normalised measure of the Euclidean discrepancy between predicted and ground-truth scores. It offers a scale-invariant assessment of prediction accuracy, making it useful for comparing performance across datasets with different score ranges. When reported, it is scaled by $\times 10^2$ for readability.

By combining these metrics—Spearman's ρ for ranking performance, (R^2) for explanatory power, MAE and RMSE for absolute error quantification and (R-ℓ_2) for normalised discrepancy—we comprehensively evaluate the performance and reliability of the model.

5.2.1 | Spearman's Rank Correlation Coefficient(ρ)

Spearman's rank correlation coefficient(ρ) is a nonparametric statistical method, which is used to measure the monotonicity between the predicted scores and the true scores, that is, whether they have a consistent increasing or decreasing trend. This is shown in Equation (13).

$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} \tag{13}$$

In the above Equation (13), d_i is the rank difference between the predicted and true values for each sample. n is the number of the samples.

5.2.2 | Relative L2 Distance (R-ℓ_2)

Unlike Spearman's rank correlation coefficient, which focuses on the ranking of test samples, the relative L2 distance measures the numerical accuracy of each sample compared to the actual value. It is formally defined as follows:

$$R\text{-}\ell_2 = \frac{1}{N}\sum_{n=1}^{N}\left(\frac{|s_n - \hat{s}_n|}{s_{\max} - s_{\min}}\right)^2 \tag{14}$$

In the above Equation (14), $R\text{-}\ell_2$ represents the relative L2 distance. N represents the total number of samples; s_n represents the ground truth score of the n th sample; $\hat{s}_n$ represents the predicted score of the n th sample; and $s_{\max}$ and $s_{\min}$ represent the maximum and minimum ground truth scores, respectively.

5.2.3 | Coefficient of Determination (R^2)

The coefficient of determination is the proportion of the variability explained by the model to the total variability, which is used to assess the explanatory power of the model prediction. It measures the model's explanatory power by comparing the predicted values with the true values. The coefficient of determination R^2 ranges from 0 to 1. The closer R^2 is to 1, the better the model fits the data. The formula for R^2 is shown in the following Equation (15):

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} \tag{15}$$

In the above formula, SS_{res} is the residual sum of squares (the sum of the squared differences between the predicted values and the true values). SS_{tot} is the total sum of squares (the sum of the squared differences between each true value and its mean).

5.2.4 | Mean Absolute Error (MAE)

MAE is the average of the absolute values of the prediction errors. It is used to measure the mean of the absolute differences between the predicted value and the true value, which is shown in Equation (16):

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{16}$$

5.2.5 | Root Mean Square Error (RMSE)

It is used to measure the root mean square difference between the predicted value and the true value. RMSE involves the errors of all data points rather than individual extreme values. This makes RMSE a robust error measure that can reflect the overall prediction performance as shown in Equation (17):

$$\text{RMSE} = \frac{1}{n}\sum_{i=1}^{n}\frac{\left|y_i - \hat{y}_i\right|}{y_i} \tag{17}$$

In Formulas (16) and (17), y_i is the true value, $\hat{y}_i$ is the predicted value and n is the number of samples.

5.3 | Experiment and Result Analysis

In this experiment, the ratio of training set to test set is 3:1, the training batch size is 4 and test batch size is 4. The number of training iterations is 100. The Adam optimiser is adopted as the gradient optimisation algorithm in this experiment. Adam combines the momentum method with an adaptive learning rate, allowing it to automatically adjust the learning rate for each parameter and use momentum to accelerate model convergence. The number of multi-head attention is set as 4.

5.3.1 | Result Analysis

To evaluate the effectiveness of our proposed method on the TaiChi-AQA dataset, we compare it against several state-of-the-art action quality assessment (AQA) approaches in the computer vision domain. These include USDL [18], DAE [19], HGCN [21], CoRe [20], T2CR [30], TSA [5], TPT [22] and FineParser [35]. The comparison is conducted under the 40-point experimental setting with results summarised in Table 6.

Compared with uncertainty modelling methods (USDL [18], DAE [19], HGCN [21]), our method achieves significant improvements in multiple indicators. For instance, compared with the DAE model, the Spearman's rank correlation coefficient(ρ) has improved by 1%, indicating that our method is better at predicting the ranking of scores, which is highly consistent with the ranking of the actual scores. The coefficient of determination R^2 has increased by 1.6%, indicating that our model performs better in explaining the variation in the target scores and the prediction results are closer to the actual scores. In addition, the mean absolute error (MAE) is reduced from 0.7952 to 0.6795, the root

TABLE 6 | Comparison of the 40-point experimental data. Bold values indicate superior performance achieved by our method compared to competing approaches.

Method	SRCC(ρ)	R^2	MAE	RMSE	R-ℓ_2 ($\times 10^2$)
USDL [18]	0.9466	0.9127	0.8463	1.1437	0.4871
DAE [19]	0.9477	0.9206	0.7952	1.0907	0.4592
HGCN [21]	0.9509	0.8933	0.9735	1.2646	0.3633
CoRe [20]	0.9519	0.9242	0.7180	1.0657	0.4121
T^2CR [30]	0.9527	0.9112	0.8152	1.1538	0.4493
TSA [5]	0.9503	0.9079	0.8855	1.1749	0.5010
TPT [22]	0.9454	0.9011	0.8174	1.1292	0.4438
FinePaser [35]	0.9545	0.9255	0.7214	1.0460	0.3970
Ours	**0.9574**	**0.9364**	**0.6795**	**0.9761**	**0.3581**

mean square error (RMSE) is reduced from 1.0907 to 0.9761 and the R-ℓ_2 is reduced from 0.4592 to 0.3581, indicating that the error is significantly reduced. Our method integrates a multi-head attention mechanism to capture local pose details and global motion coherence. It uses a gating mechanism to suppress noisy features, reducing the distribution bias and absolute prediction error compared to the USDL and DAE.

CoRe [20] and T^2CR [30] are contrastive regression methods, which improve AQA performance by optimising inter-sample contrastive loss. Between these two methods, CoRe achieves better error metrics (MAE = 0.7180, RMSE = 1.0657), and T^2 achieves a higher ρ (0.9527). However, our method still outperforms both in all indicators: compared with CoRe, our method improves ρ by 0.55%, R^2 by 1.22%, MAE by 3.85%, RMSE by 8.96% and R-ℓ_2 by 5.4%; compared with T^2CR, our method improves ρ by 0.47%, R^2 by 2.52%, MAE by 13.57%, RMSE by 17.77% and R-ℓ_2 by 9.12%. The limitation of contrastive regression methods is that they focus on distinguishing score differences between samples but ignore the fine-grained spatiotemporal cues of TaiChi itself. For example, CoRe's group-aware contrastive loss and T^2CR's two-path target-aware mechanism prioritise inter-sample discrimination over intra-sample movement details. Our method distinguishes between high and low-quality samples and also optimises the modelling of key postures, thus achieving both strong ranking correlation and low absolute error.

Comparison with spatiotemporal parsing methods (TSA [5], TPT [22], FineParser [35]), TSA emphasises procedure-aware assessment, TPT uses temporal parsing transformers to model time-series features, and FineParser proposes a fine-grained spatiotemporal parser for human-centric actions. Among them, FineParser achieves the closest performance to our method with a ρ of 0.9545, R^2 of 0.9123, MAE of 0.7422 and RMSE of 1.0324. However, our method still outperforms FineParser by 0.29% in ρ, 2.41% in R^2, 8.45% in MAE, 5.45% in RMSE and 9.43% in R-ℓ_2. TSA, which is not tailored for TaiChi, performs the worst: our method surpasses it by 1.99% in ρ, 7.64% in R^2, 38.80% in MAE, 32.59% in RMSE and 53.00% in R-ℓ_2. TPT, which focuses on temporal parsing but lacks uncertainty modelling, lags behind our method by 1.20% in ρ, 3.53% in R^2, 16.87% in MAE and 13.47% in RMSE. The advantage of our method over this group stems from its ability to capture subtle differences in actions and predict the distribution of scores rather than just predicting scores.

Across all three categories of methods, our method consistently achieves the best performance in all evaluation metrics validating its comprehensive advantages in TaiChi-AQA. By targeting the unique characteristics of TaiChi—subtle posture differences and continuous movement fluency—our method overcomes the limitations of existing approaches. These results demonstrate that our approach significantly reduces prediction errors and offers higher regression prediction accuracy, while also excelling in correlation, fitting ability and error control compared to other methods.

To further validate the effectiveness of our proposed Multi-Head Gated Attention Module (MHGM), we compare it with three state-of-the-art attention-based modules—MSCA [46], External-Attention (EA) [47] and ECA [48]—all integrated with the same I3D feature extractor as the baseline. The comparative results under the 40-point experimental setting are summarised in Table 7 where our I3D+MHGM method achieves the best performance across all evaluation metrics, demonstrating the superiority of our module in modelling action quality features.

In terms of SRCC, our method reaches 0.9574, which is 0.44% higher than I3D+MSCA (0.9530) and 0.90% higher than I3D+ECA (0.9484). For the coefficient of determination (R2), our method achieves 0.9364, outperforming I3D+MSCA (0.9290) by 0.74% and I3D+ECA (0.9222) by 1.42%. These results demonstrate that MHGM better captures the monotonic relationship between predicted scores and ground truth and explains more variance in the actual quality scores. For error quantification, our method yields the lowest MAE (0.6795), RMSE (0.9761) and R-ℓ_2(0.3581). Compared to I3D+MSCA, this represents a 3.01% reduction in MAE, 5.53% reduction in RMSE and 5.27% reduction in R-ℓ_2; against I3D+ECA, the improvements reach 7.69%, 10.37% and 12.05%, respectively. These consistent decreases indicate MHGM reduces average errors, large deviations and scale-dependent errors owing to its ability to capture long-term dependencies in Tai Chi movements, noise suppression via gating mechanisms and uncertainty modelling of the score autoencoder.

A more intuitive comparison line chart of 100 experimental results is shown in Figure 8 where our method outperforms others across all indicators with significant robustness and stability. In the early stage of training, the indicator values of our method rise rapidly reflecting a higher convergence speed. In the later stage, the indicator curve fluctuates less, which further proves its stability. In terms of ρ-value and R^2 indicators, our method shows higher correlation and goodness of fit, indicating a closer relationship between the predicted results and the actual values. In terms of MAE and RMSE metrics, our method has lower errors, indicating that its prediction accuracy is significantly better than that of others.

In addition, the score range of the movement quality module in Tai Chi teaching is fixed at 0–5 points. Therefore, we also conduct an experiment where the Final Score is directly obtained from the addition of the exercise level module and the action quality module. The final score ranges from 0 to 25 points. The experimental results are recorded in Table 8, which further validates the superiority of the proposed method.

Since the predicted score in this paper is generated from a Gaussian distribution, the sample scores are more concentrated around the mean when the Final Score is in the range of 0–25 points. As a result, the predicted scores are closer to the actual scores. Notably, all four performance indicators (ρ, R^2, MAE, RMSE, R-ℓ_2) achieve better results when the score range is 0–25 compared to 0–40.

5.3.2 | Ablation Study

To clarify the individual contribution of each proposed component, we conduct a comprehensive ablation study on the TaiChi-AQA dataset. All variants share the same I3D feature extractor and training protocol differing only in the investigated

TABLE 7 | Comparative experiments with other modules. Bold values indicate superior performance achieved by our method compared to competing approaches.

Method	SRCC(ρ)	R^2	MAE	RMSE	R-ℓ_2 ($\times 10^2$)
I3D+MSCA [46]	0.9530	0.9290	0.7096	1.0314	0.4108
I3D+EA [47]	0.9491	0.9232	0.7193	1.0726	0.4641
I3D+ECA [48]	0.9484	0.9222	0.7564	1.0798	0.4786
I3D+MHGM (Ours)	**0.9574**	**0.9364**	**0.6795**	**0.9761**	**0.3581**

Note: I3D indicates the baseline I3D feature extractor.

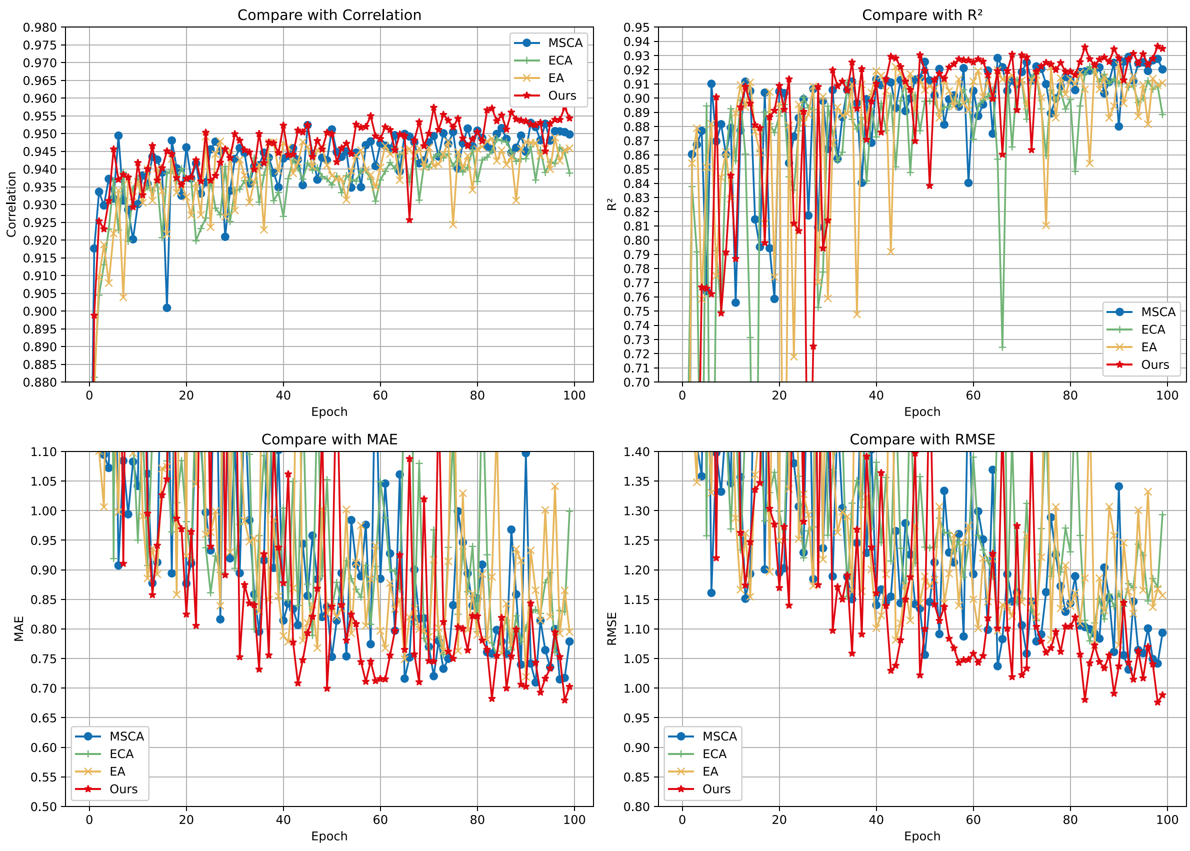

FIGURE 8 | Comparison line chart of test data.

module. Quantitative results are reported in Table 9. The following will conduct an ablation analysis from the perspectives of the loss function, the gating mechanism and the score distributed autoencoder.

5.3.2.1 | Loss Function. Replacing our full loss function (Equation 12), which includes both the mean squared error between the predicted score and the ground truth as well as an additional term to align the model's predicted mean $\mu(x)$ with the ground truth, with a vanilla mean squared error (MSE) loss (only use the predicted score and the ground truth) leads to mixed but overall degraded performance. Specifically, SRCC decreases from 0.9574 to 0.9558 (a relative drop of approximately 0.17%), R^2 drops from 0.9364 to 0.9315 and R-ℓ_2 increases from 0.3581 to 0.4003 (an absolute rise of 0.0422, indicating worse relative error). Although MAE and RMSE show slight improvements (MAE from 0.6795 to 0.6492, RMSE from 0.9761 to 0.9631), these gains come at the expense of correlation-based metrics, which are crucial for action quality assessment. This underscores the value of the additional term in enhancing model stability by ensuring the predicted mean remains consistent with actual scores.

5.3.2.2 | Gated Mechanism. Ablating the gated multilayer perceptron (gMLP) and reverting to a standard multilayer perceptron (MLP) results in performance degradation across most metrics: SRCC decreases from 0.9574 to 0.9508 (a relative drop of approximately 0.69%), R^2 drops from 0.9364 to 0.9267, MAE increases from 0.6795 to 0.7137 (an absolute rise of 0.0342), RMSE rises from 0.9761 to 1.0478 (an absolute increase of 0.0717) and R-ℓ_2 worsens from 0.3581 to 0.3985 (an absolute rise of 0.0404). These changes highlight the gated mechanism's importance in dynamically modulating feature representations. In action quality assessment tasks such as those in TaiChi-AQA where videos often contain noisy elements such as variable lighting, background interference or inconsistent motion patterns, the gating enables selective emphasis on relevant features while attenuating distractions leading to more robust and accurate score predictions overall.

5.3.2.3 | Score Distributed Autoencoder (DAE). Substituting the probabilistic DAE with a direct regression head results in performance degradation across all metrics: SRCC decreases from 0.9574 to 0.9482 (a relative drop of approximately 0.96%), R^2 drops from 0.9364 to 0.9262, MAE increases from 0.6795 to 0.7324, RMSE rises from 0.9761 to 1.1436 and R-ℓ_2 worsens from 0.3581 to 0.4312 (an absolute rise of 0.0731). The DAE better handles inherent uncertainties in subjective annotations, such as inter-rater variations in TaiChi-AQA, leading to more robust generalisation and reduced overfitting compared to deterministic regression.

In summary, these ablations demonstrate that each component contributes meaningfully to the model's efficacy, collectively addressing key challenges in video-based AQA such as feature noise, label uncertainty and regression calibration.

6 | Conclusion

In this paper, we propose a method for action quality assessment (AQA) based on the 24-posture Tai Chi dataset. We utilise I3D to extract the spatiotemporal features of action videos, which are then input into the multi-head attention mechanism

TABLE 8 | Comparison of the 25-point experimental data. Bold values indicate superior performance achieved by our method compared to competing approaches.

Method	SRCC(ρ)	R^2	MAE	RMSE	R-ℓ_2 ($\times 10^2$)
USDL [18]	0.9533	0.9224	0.5811	0.7899	0.5593
DAE [19]	0.9607	0.9382	0.4709	0.7049	0.4462
HGCN [21]	0.9625	0.9314	0.5188	0.7427	0.4266
CoRe [20]	0.9601	0.9329	0.4785	0.7349	0.4546
T^2CR [30]	0.9629	0.9362	0.4739	0.7166	0.4322
TSA [5]	0.9546	0.9157	0.5564	0.8237	0.5710
TPT [22]	0.9607	0.9287	0.4822	0.7421	0.4624
FinePaser [35]	0.9619	0.9415	0.4264	0.6863	0.3964
Ours	**0.9652**	**0.9419**	**0.4064**	**0.6834**	**0.3788**

TABLE 9 | Ablation study on TaiChi-AQA.

Method	SRCC(ρ)↑	R^2↑	MAE↓	RMSE↓	R-ℓ_2 ($\times 10^2$)↓
Full model	**0.9574**	**0.9364**	**0.6795**	**0.9761**	**0.3581**
w/o MHGM	0.9477	0.9206	0.7952	1.0907	0.4592
w/o gated MLP	0.9508	0.9267	0.7137	1.0478	0.3985
w/o DAE (regression head only)	0.9482	0.9262	0.7324	1.1436	0.4312
w/o full loss (Equation 12)	0.9558	0.9315	0.6492	0.9631	0.4003

Note: Bold numbers indicate the best performance. ↑: higher is better, ↓: lower is better.

and gMLP module for feature learning. The learnt features are subsequently mapped to a Gaussian distribution using a score distribution autoencoder (DAE), enabling the model to capture subtle differences in action sequences and prioritise key information that significantly impacts the score. This approach demonstrates a significant improvement in the accuracy of the assessment. Experimental results confirm that this method effectively handles the uncertainty and variability inherent in Tai Chi actions.

In the future, we aim to further optimise the method for action quality assessment on the 24-posture Tai Chi dataset. Integrating multimodal data, such as physiological signals and motion capture, and performing comprehensive analyses will enhance the accuracy and depth of Tai Chi action assessments. We hope this research will not only advance the understanding and evaluation of Tai Chi but also provide a novel perspective for the modernisation of traditional cultural practices.

Author Contributions

Dejin Wang: funding acquisition, project administration, resources, validation, writing – review and editing. **Fengyan Lin:** data curation, investigation, writing – review and editing. **Kexin Zhu:** conceptualization, methodology, supervision. **Zhide Chen:** resources, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request. The video data used in this study were sourced from publicly available platforms. The video features extracted using the I3D model will be made publicly available for further research.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting Information S1: cvi270053-sup-0001-suppl-data.pdf.

Appendix A: Action Category

TABLE A1 | Action category corresponding abbreviation table.

Action category abbreviation	Action category full name
CF1	Commencing form
PWM	Part the wild Horse's Mane on both sides
WCSW	White crane spreads its wings
BKTS	Brush knee and twist step on both sides
PP	Play pipa
RM	Repulse monkey
GBT-L	Grasp the bird's tail-left
GBT-R	Grasp the bird's tail-right
SW1	Single whip 1
WHLC	Wave hands like clouds
SW2	Single whip 2
HPH	High pat on Horse
KRH	Kick with right heel
SOEBF	Strike opponent's ears with both fists
TKLH	Turn and kick with left heel
LSCD	Left snake creeps down and golden rooster stands on one leg
RSCD	Right snake creeps down and golden rooster stands on one leg
JLWS	Jade lady weaves shuttles
NSB	Needle at sea bottom
FA	Flash the arm
TDPP	Turn, deflect downward, Parry and Punch
AC	Apparent close up
CH	Cross hands
CF2	Closing form